Random Does Not Mean the Same

AP Statistics · Topic 3.3 (CED 1.11) · B: 2026-09-29 · E: 2026-10-07 · TEACHER KEY

CED TETHER

- **Skill 1.C** — Describe an appropriate method for gathering and representing data.
- **EU DAT-2** — The way we collect data influences what we can and cannot say about a population.
- **LO DAT-2.C** — Identify a sampling method, given a description of a study. [Skill 1.C]
- **EK DAT-2.C.1** — When an item from a population can be selected only once, this is called sampling without replacement. When an item from the population can be selected more than once, this is called sampling with replacement.
- **EK DAT-2.C.2** — A simple random sample (SRS) is a sample in which every group of a given size has an equal chance of being chosen. Mechanisms include numbering individuals and using a random number generator, a table of random numbers, or drawing a card from a deck without replacement.
- **EK DAT-2.C.3** — A stratified random sample involves dividing a population into separate groups called strata based on shared attributes (homogeneous grouping). Within each stratum a simple random sample is selected, and the selected units are combined to form the sample.
- **EK DAT-2.C.4** — A cluster sample involves dividing a population into smaller groups called clusters, ideally with heterogeneity within each cluster and similarity among clusters. A simple random sample of clusters is selected, and data are collected from all observations in the selected clusters.
- **EK DAT-2.C.5** — A systematic random sample selects members from a population according to a random starting point and a fixed, periodic interval.
- **EK DAT-2.C.6** — A census selects all items/subjects in a population.
- **LO DAT-2.D** — Explain why a particular sampling method is or is not appropriate for a given situation. [Skill 1.C]
- **EK DAT-2.D.1** — There are advantages and disadvantages for each sampling method depending upon the question to be answered and the population from which the sample will be drawn.

Essential question: Which random sampling method best matches the structure of the population and the question?

DOK-3 skill: 4.A · **FRQ pattern:** `srs-vs-systematic-vs-stratified-choose-and-defend`

DOK rationale: Part (c) coordinates the population structure, the guarantee supplied by stratification, and the ordering risk in systematic sampling to choose and defend a method rather than merely identify it.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to compare any two methods before naming them.
- Begin the video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and ask what each method guarantees.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Commit to whether the methods are equivalent and cite one difference.

- Complete the video follow-along about SRS, systematic, stratified, and cluster sampling.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- For an SRS, what must have an equal chance: each person or each group of 50?
- Which method forces every grade to appear?
- Could an SRS contain many students from one grade by chance?
- What must be true about the roster order for every 24th name to create a repeated type?

Adult role

Solo teacher. Circulate 5 pairs. Ask for the guarantee and the failure mode before accepting a method name; do not treat all chance-based designs as interchangeable.

[WARN] WATCH FOR

- Using random start as proof that the systematic sample is an SRS.
- Confusing strata, sampled within every group, with clusters, where whole selected groups are measured.
- Claiming any sorted list biases systematic sampling without identifying periodic alignment.

SCORING PART (c) — E / P / I

Expected elements

- **chooses-stratified-for-grade-differences** (required) — Chooses Method 3 and ties the choice to substantial grade differences in sleep
- **states-representation-guarantee** (required) — Explains that random sampling within every grade guarantees each grade appears in the sample
- **contrasts-other-methods** (required) — States that the SRS and systematic plans do not impose a required count from each grade
- **explains-periodicity-risk** (required) — Explains how a roster pattern aligned with the interval could repeatedly select one program type and omit others

E	Chooses Method 3 for the stated goal, cites its every-grade guarantee, contrasts both alternatives, and explains the interval-ordering risk precisely.
P	Chooses Method 3 with a correct grade-based reason but incompletely contrasts the alternatives or says only that roster order could matter without explaining the repeated-position mechanism.
I	Calls all three methods SRSs, chooses solely because one sounds more random, or claims every systematic sample is biased without reference to ordering.

Common mistakes

- Calling Method 1 an SRS because its starting point is random
- Calling Method 3 a cluster sample instead of sampling within every stratum
- Saying an SRS guarantees exactly equal numbers from each grade
- Claiming alphabetical or grade ordering by itself proves bias without an interval-aligned pattern

[STAR] TODAY'S PROBLEM — annotated

Suppose a school of 1,200 students wants to estimate nightly hours of sleep. Three teams propose samples of 50.

Method 1: Choose a random start among the first 24 names on the alphabetical roster, then take every 24th name.

Method 2: Put all 1,200 names in a hat, mix thoroughly, and draw 50 without replacement.

Method 3: Separate the roster by grade, then randomly choose 12 or 13 students from each grade, for 50 total.

A student says, *They all use randomness, so they are the same.*

Three proposed samples

#	Rule	n	Per grade
1	Every 24th	50	Not fixed
2	Hat	50	Not fixed
3	Within grade	50	12 or 13

FIRST TAKE (5 min)

Are all three methods the same kind of random sample? Choose one comparison and explain your instinct.
Key: Not graded. Expect the word random to make the three methods look equivalent at first.

Rules callout "THREE RANDOM METHODS (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Name each method: systematic random, simple random, or stratified random.

Key: Method 1 is systematic random sampling. Method 2 is a simple random sample. Method 3 is stratified random sampling, with grade as the stratum.

DOK 2 **(b)** Which method is an SRS of 50 students? Explain using what must have an equal chance, not just the word random.

Key: Method 2 is the SRS because every possible group of 50 students has an equal chance to be drawn. Method 1 allows only groups generated by one of 24 starts, and Method 3 forces 12 or 13 students from each grade, so neither gives every group of 50 an equal chance.

DOK 3 **(c)** If sleep differs substantially by grade, choose one method and defend it with the feature that settles your choice. Explain what the other two methods fail to guarantee. Then suppose the roster has a repeating 24-name pattern tied to a school program; explain the specific risk for Method 1.

Key: Use Method 3 when grade is strongly related to sleep because randomly sampling within every grade guarantees that every grade is represented. Method 2 can include very unequal grade counts by chance; Method 1 also does not set a required count within each grade. If a 24-name roster cycle is tied to a program and every 24th position has the same program status, Method 1 can repeatedly select that type of student and miss others. A grade-sorted roster alone is not automatically biased; the danger is an ordering pattern that lines up with the interval.

EXIT

One thing the video changed about my first take: ______________________________